

Does Continued Pretraining on a Learner Corpus Improve Automated Essay Scoring on English Proficiency Tests? Evidence from EFCAMDAT

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Abstract

Recent automated essay scoring (AES) studies on English proficiency assessments often relied on pretrained transformer models. However, because these models are typically pretrained on native English corpora, they may be less effective at capturing the linguistic characteristics of L2 learner writing. In this study, we investigate whether continued pretraining on an L2 English learner corpus can improve the performance of a pretrained transformer for AES. We perform domain-adaptive continued pretraining on the EFCAMDAT corpus using BERT as the base model. We then conduct two experiments, using two different essay datasets drawn from standardized English proficiency tests, namely FCE and IELTS. The first experiment compares the domain-adapted model against several baselines, including its original base checkpoint, on the target datasets. The second evaluates few-shot cross-dataset transfer to examine whether continued pretraining reduces transfer-induced performance degradation relative to the base checkpoint. Our results show that the domain-adapted BERT model performed worse than both its original base checkpoint and the other baseline models. However, under our cross-dataset transfer setting, the domain-adapted model showed a smaller transfer-induced performance drop than base BERT, with improvements in RMSE in both transfer directions and additional gains in Spearman and Pearson when IELTS was the target dataset.

1 Introduction

English proficiency assessment plays an important role in educational placement, certification, and language learning, with writing being a key component because it reflects a learner’s ability to organize ideas and communicate effectively in English. However, evaluating writing is resource-intensive and time-consuming, especially when large numbers of responses must be assessed consistently. Therefore, Automated Essay Scoring (AES) has

clear practical value in English proficiency settings, where it can support efficient and timely evaluation of learner writing.

Despite this practical importance, AES for English proficiency writing remains challenging. Unlike many general essay datasets, English assessment essays are produced by L2 learners, which often contain distinctive lexical, grammatical, and discourse patterns associated with developing language ability. Yet recent transformer-based AES approaches commonly rely on general-domain pretrained language models (Qiu et al., 2024; Schmalz and Brutti, 2021), raising the question of whether models adapted to learner writing may be better suited to this setting .

Beyond the challenge of modeling learner writing itself, AES for English proficiency assessment must also contend with variation across tests. Currently, a wide range of standardized English proficiency tests are used worldwide, such as IELTS, TOEFL, and Cambridge English examinations, each of which differs in task type, prompt design, scoring criteria, and assessment context. This diversity creates a practical challenge for AES, as a model developed for one test may not perform reliably on another. Building a separate AES system for every individual test would be both time-consuming and inefficient, especially as new test formats and writing tasks continue to emerge. A more practical direction is therefore to develop an AES approach with stronger transferability, so that one model can generalize more effectively across different English proficiency assessments.

In this study, we conduct domain-adaptive continued pretraining on BERT using the EFCAMDAT corpus, a large-scale English L2 learner corpus. Our study investigates whether continued pretraining on an English learner corpus yields improved AES performance on English proficiency tests. We conduct two main experiments and one supplementary analysis, namely baseline comparisons, cross-

dataset transfer, and an ablation study. Two distinct datasets were used to ensure that the evaluation was not limited to a single assessment context and to enable cross-dataset transfer experiments examining whether the proposed approach can mitigate the performance degradation caused by dataset shift

We hypothesize that continued pretraining on a learner corpus improves both in-domain and cross-dataset AES performance by shifting the model’s parameters closer to the target domain, namely the linguistic distribution of English learner writing. Taken together, our research aims to answer the following questions:

RQ1 Does continued pretraining on a learner corpus improve AES performance on English proficiency tests compared to general-domain pretrained baselines?

RQ2 Does continued pretraining on a learner corpus reduce performance drop when transferring from one English proficiency test dataset to another in a few-shot setting?

The remainder of this paper is organized as follows. Section 2 reviews the relevant background and related work. Section 3 presents the experimental setup, including the data, preprocessing procedures, domain-adaptive pretraining process, experimental design, and evaluation protocol. Section 4 reports the results of the baseline comparison, cross-dataset transfer, and ablation study, and discusses their implications. Finally, Section 5 concludes the paper and outlines directions for future work.

2 Background and Related Work

Automated Essay Scoring on English proficiency tests has been the topic of research for over 20 years. Early AES studies focusing on this specific assessment setting often relied on classical machine learning techniques and feature engineering. For example, [Burstein and Chodorow \(1999\)](#) used e-rater, an AES system that provides score estimations based on a wide range of engineered linguistic features, such as syntactic structures and variety, discourse Cues, and the organization of ideas, to score Test of Written English (TWE) essays written by non-native English speakers. While these early studies demonstrated that AES could be effectively applied to learner writing, their approaches were typically developed and evaluated

within relatively fixed assessment settings, where the prompts for essays are static.

Since the emergence of the Transformer architecture, AES research has shifted toward language models. In the context of English proficiency assessment, several studies have explored models such as BERT and DeBERTa for scoring learner essays ([Qiu et al., 2024](#); [Schmalz and Brutti, 2021](#)). However, most of this work relies on general-domain transformers pretrained on corpora such as BookCorpus and Wikipedia, which consist largely of native-written text ([Devlin et al., 2019](#)). This creates a mismatch between the pretraining data and the target domain, since learner writing often contains grammatical errors, awkward phrasing, and unconventional lexical usage. As a result, general-domain language models may be less effective at capturing the characteristics of learner language, which can in turn limit AES performance and robustness.

A common approach to addressing distribution mismatch is domain adaptation. In AES, domain adaptation has been explored previously, but existing studies have focused largely on addressing the problem of prompt mismatch, where models suffer performance degradation when scoring essays written in response to unseen prompts ([Phandi et al., 2015](#); [Cao et al., 2020](#); [Jiang et al., 2023](#)). While prompt-level domain adaptation has proven effective in such settings, it is less suitable for AES in English proficiency tests, where there is no fixed prompt because the prompts are intentionally designed to vary across different topics and task types to effectively assess the language proficiency of the test taker. In this context, the distribution of learner language itself becomes a more appropriate target for adaptation.

A common domain adaptation approach widely used with Transformer-based models is domain-adaptive pretraining (DAPT) ([Gururangan et al., 2020](#)). In this method, a pretrained language model undergoes further pretraining on text drawn from the target domain. In the context of computer-assisted language learning, [Stearns et al. \(2024\)](#) applied DAPT to the BERT model using the EFCAMDAT corpus, a dataset consisting of learners’ writings from different levels of proficiency. Their findings demonstrate the promise of this approach, as the adapted model achieved strong results on intrinsic evaluations. However, the study did not examine whether these gains transfer to downstream tasks through task-specific fine-tuning.

In summary, the literature suggests a clear gap. Previous transformer-based AES studies on English proficiency tests have mainly relied on general-domain models, despite the mismatch between pretraining corpora and learner writing. While domain adaptation has been employed in AES, prior work has applied this technique at the level of prompts, which is less suitable for English proficiency tests, where prompts vary substantially across topics, task types, and test administrations. These limitations motivate research toward a more stable adaptation target, namely, the linguistic characteristics of English L2 learner writing. Although domain adaptation toward learner-language distributions has been explored, it has received limited attention in automated essay scoring.

Our study addresses these gaps by investigating whether a transformer-based AES model benefits from continued pretraining on a learner corpus in terms of in-domain scoring accuracy and cross-dataset robustness in English proficiency assessment.

3 Experimental Setup

3.1 Data

3.1.1 Pretraining Corpus

EFCAMDAT. The EFCAMDAT is an open-access corpus of L2 English learner writings, collected from submissions to Englishtown, the online language school of EF Education First (Huang et al., 2017; Geertzen et al., 2014). The corpus comprises texts written by learners from diverse demographic backgrounds, representing 198 nationalities, and spans proficiency levels from CEFR A1 to C1. The second release of the EFCAMDAT corpus comprises 1,180,310 texts written by 174,743 learners, totaling 83,543,480 word tokens.

In this study, we use the cleaned version of the EFCAMDAT dataset for continued pretraining (Shatz, 2020). This variant underwent several cleaning steps that removed markup tags and text dominated by markup tags, ultra-short scripts, texts with substantial non-English content, duplicates across documents, and extreme-length outliers. The resulting sub-corpus comprises 45,390,000 word tokens.

3.1.2 Downstream datasets

CLC FCE Dataset. The CLC FCE dataset is provided by the Cambridge Learner Corpus (CLC) as a part of a collaborative project between Cambridge

University Press and Cambridge Assessment (Yannakoudakis et al., 2011). It contains scripts written by upper-intermediate English as a Second or Other Language (ESOL) learners taking the First Certificate in English (FCE) examination. Each exam script contains two responses to the two tasks in the FCE writing section, which asks candidates to write a letter, a report, an article, a composition, or a short story between 200 and 400 words. In this dataset, each text is labeled with two different score annotations: an overall score that the script’s author achieved across both writing tasks, and a score assigned to that specific response. In this study, we use the latter score as the target variable when fine-tuning and evaluating downstream performance.

IELTS Writing Scored Essays. The IELTS Writing Scored Essays is a publicly available dataset on Kaggle¹. This dataset was previously utilized by Liew and Tan (2025) and Qiu et al. (2024) in studies on the use of Large Language Model in AES, making it a good fit for our study. The dataset contains 1274 unique essays, covering both IELTS Writing Task 1 and Task 2, each annotated with a score ranging from 0 to 9. However, in this study, we only use IELTS Task 2 essays in the experiments. Task 1 essays were excluded because the dataset did not contain the figures or diagrams referenced in the prompts, making these responses incomplete for reliable modeling. As a result, the final dataset used in our experiments consists of 710 Task 2 essays.

3.2 Preprocessing

For the IELTS dataset, samples with missing essay texts or labels were removed. Duplicate entries were also excluded, along with ultra-short essays containing fewer than 100 characters. In addition, samples containing foreign language text were also discarded to improve data consistency.

The public FCE dataset uses a single holistic score ranging from 0,0 to 5,3, which later AES work often converts to a 0–20 integer scale (Gaudeau, 2025; Zhang et al., 2018). In our study, we adopt the conversion scheme of Zhang et al. (2018) to map the original FCE scores to this 0–20 scale.

¹<https://www.kaggle.com/datasets/mazlumi/ielts-writing-scored-essays-dataset>

3.3 Domain-adaptive Pretraining

In this study, we use BERT (Devlin et al., 2019) as the model checkpoint for continued pretraining on the EFCAMDAT corpus. We selected BERT because it has been previously used in AES research, making it a suitable and well-established foundation for our study (Wang et al., 2022). In addition, Stearns et al. (2024) have previously explored domain-adaptive continued pretraining on BERT using the EFCAMDAT corpus, which reported positive gains in intrinsic evaluation relative to the original BERT checkpoint. This motivates us to examine whether such benefits also transfer to downstream AES on English proficiency assessment essays. Following Stearns et al. (2024), we conduct continued pretraining with a Masked Language Modeling objective, masking 15% of WordPiece tokens in the training data and training the model to predict the masked items from their surrounding context. Their study also mentions filtering out texts with more than 512 WordPiece tokens, which we also follow.

3.4 Experiments

To answer our research questions, we conduct two main experiments and one supplementary analysis: baseline comparison, cross-dataset transfer, and a proficiency-based ablation study.

Baseline Comparison. In the baseline comparison, the domain-adapted model is fine-tuned and evaluated separately on the FCE and IELTS datasets, and its performance is compared against other baseline transformer models trained under the same downstream settings. In addition to BERT, we selected RoBERTa (Liu et al., 2019), and DistilBERT (Sanh et al., 2020) as baseline models because they have been successfully implemented in prior AES research (Gui, 2025; Qiu et al., 2024). Specifically, the baseline checkpoints used in our experiments are google-bert/bert-base-uncased, FacebookAI/roberta-base, and distilbert/distilbert-base-uncased. We also initially included DeBERTa (He et al., 2021) in the baseline setup. However, it consistently collapsed to constant predictions during fine-tuning. As we were unable to determine the underlying cause of this behavior within the scope of the study, we excluded its results from the final comparison. Figure 1 visualizes this experiment.

Cross-dataset Transfer. In this experiment, we assess transferability in both directions: from IELTS to FCE and vice versa. For each direction, the model is first fine-tuned on the source dataset, then subjected to an additional few-shot adaptation stage on the target dataset using 50 training samples, before being evaluated on the target test set. Because the two datasets use different scoring ranges, the task-specific scoring head is reinitialized rather than reused from the source task. We then compare the performance degradation associated with cross-dataset transfer between the domain-adapted model and the base BERT model. This allows us to investigate whether continued pretraining on a learner corpus helps mitigate the performance degradation caused by cross-dataset transfer. Figure 2 demonstrates this experiment.

Ablation Study. We divide the EFCAMDAT corpus by learner proficiency levels and train separate variants on selected proficiency ranges on the CEFR scale, including A1–A2 and B1–B2–C1, in order to examine whether the proficiency range of the adaptation corpus influences downstream AES performance. To ensure a fair comparison across ablation settings, we controlled for training volume by capping the maximum number of training tokens. This prevents differences in token count from acting as a confounding factor in the observed performance differences, especially given the substantially larger number of A1–A2 texts in EFCAMDAT relative to the higher proficiency levels. The maximum number of training tokens was capped at 21,019,615, corresponding to the total number of tokens available in the B1–B2–C1 variant, which accounts for approximately 35% of the total WordPiece tokens in the EFCAMDAT.

3.5 Evaluation Protocol and Metrics

In our experiments, both downstream datasets are split with a 70/15/15 ratio for training, validating, and evaluation. For the FCE dataset in particular, we split by the provided author ID instead of individual text. This prevents essays from the same learner from appearing in both training and evaluation sets, thereby reducing the risk of author-specific leakage and yielding a more reliable estimate of generalization. For all experiments, final performance is reported on the held-out test set that is not used during training or model selection. To ensure fairness, all experiments are conducted on the NVIDIA L4 GPU using the same hyperparam-

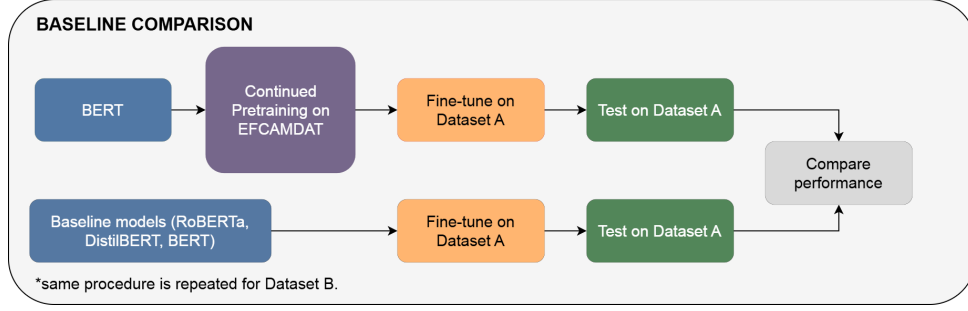


Figure 1: Baseline comparison setup. The domain-adapted model is obtained by continuing pretraining from the BERT base checkpoint on EFCAMDAT, then fine-tuned and evaluated on each downstream dataset. Its performance is compared against standard pretrained baselines trained and evaluated under the same downstream setup. The same procedure is repeated for both downstream datasets.

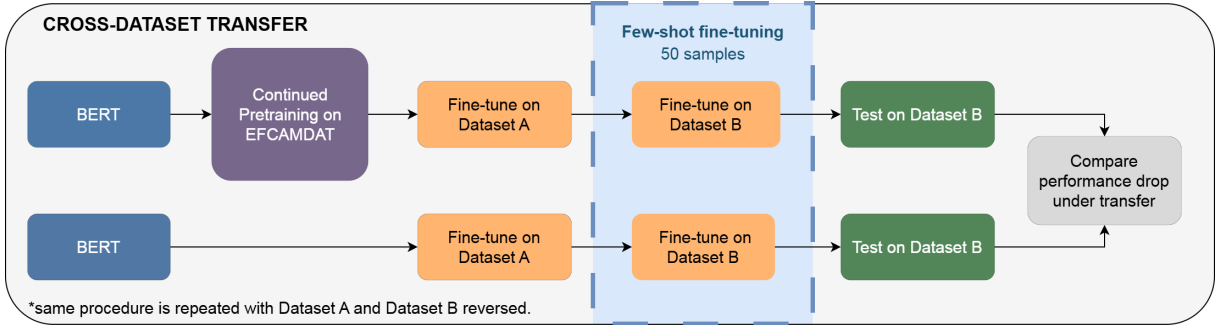


Figure 2: Cross-dataset transfer setup. Models are first fine-tuned on a source dataset, then further adapted to the target dataset using a few-shot fine-tuning stage with 50 training samples, and finally evaluated on the target test set. Transfer-induced performance drop is compared between the domain-adapted model and the BERT baseline. The same procedure is repeated with source and target datasets reversed.

eters and a fixed seed across runs. For the detailed hyperparameters used, see Appendix A.

In this study, we conduct our experiments in a prompt-unaware setting, meaning that no explicit prompt information is provided to the model. This allows us to better isolate the effect of domain-adaptive pretraining on the linguistic characteristics of the essays themselves, rather than on prompt-specific cues. Since continued pretraining is intended to improve the model’s representation of learner writing, a prompt-unaware design provides a clearer test of whether such gains come from the essay text alone. This also provides a cleaner setting for our cross-dataset transfer experiment, where prompt mismatch can otherwise complicate the outcome.

We report RMSE, QWK, Pearson, and Spearman. Following prior AES work, we include QWK, Pearson, and Spearman because they are commonly reported metrics (Phandi et al., 2015). However, Yannakoudakis and Cummins (2015) have highlighted limitations of both correlation-based measures and kappa-based agreement coefficients for

automated text scoring (ATS) evaluation. We therefore treat RMSE as the primary metric, which is also used by Gaudeau (2025) as the main metric for validation and model selection.

4 Results and Discussion

In this section, we present and discuss the results obtained from the experiments described in Section 3.4.

4.1 Baseline Comparison

Table 1 presents the results of the baseline comparison experiment. In contrast to our hypothesis, the domain-adapted BERT model failed to outperform the baseline models and is weaker than most of them across the reported metrics. More importantly, its performance is consistently below that of the original bert-base-uncased checkpoint from which it was obtained through continued pretraining. This suggests that, in our setting, domain-adaptive pretraining on EFCAMDAT did not improve in-domain AES performance and may even have degraded the downstream usefulness of the

Model	RMSE	QWK	Sp.	Pe.
IELTS				
BERT	0.817	0.661	0.773	0.794
RoBERTa	0.872	0.462	0.741	0.654
DistilBERT	0.766	0.576	0.731	0.712
DAPT BERT	0.911	0.538	0.722	0.686
CLC FCE				
BERT	2.163	0.498	0.629	0.636
RoBERTa	2.484	0.347	0.519	0.567
DistilBERT	2.190	0.559	0.626	0.625
DAPT BERT	2.386	0.405	0.569	0.565

Table 1: In-domain AES performance on IELTS Task 2 and CLC FCE. Sp.=Spearman and Pe.=Pearson. DAPT BERT denotes bert-base-uncased further pretrained on EFCAMDAT. Best performance achieved on each dataset is shown in bold.

original pretrained representations. This provides a negative response to our **RQ1**.

Gururangan et al. (2020) show that continued pretraining is generally more effective when the adaptation corpus is closely aligned with the target task distribution. In our case, however, continued pretraining not only failed to improve performance but instead led to degradation, suggesting that the pretraining corpus may not have been sufficiently aligned with the downstream AES datasets. One possible explanation for this result is the skewed proficiency distribution of EFCAMDAT, an issue also noted by Stearns et al. (2024). Specifically, our pretraining corpus contains substantially more beginner-level texts (A1–A2) than advanced-level learner writing. In contrast, the FCE dataset consists primarily of upper-intermediate learner essays, while the IELTS dataset contains academic essays, creating a mismatch between the linguistic distribution of the pretraining corpus and that of the downstream datasets. In addition, there are other mismatches between EFCAMDAT and the downstream datasets, such as communicative purpose and genre, which may also have contributed to the observed performance degradation.

4.2 Cross-dataset Transfer

Table 2 presents the results of cross-dataset transfer in both directions. In both settings, the transferred models collapsed to near-constant predictions, resulting in QWK values of 0; accordingly, QWK is not informative in this context and is not discussed further. When FCE is the target dataset, the domain-adapted BERT model shows a clear improvement in RMSE, reducing the transfer-induced error increase by 1.427 points relative to the base BERT model, while the correlation-based metrics

remain broadly similar, with only small decreases. In the opposite direction, where IELTS is the target dataset, the domain-adapted model yields clearer evidence of improvements, consistently showing a smaller transfer-induced performance drop than the base BERT model across RMSE, Spearman, and Pearson.

These results provide qualified support for **RQ2**. Although the effect is not uniform across both transfer directions and all evaluation metrics, the domain-adapted model consistently exhibits a smaller transfer-induced performance decline than the base BERT model in RMSE. One possible interpretation is that domain-adaptive pretraining may encourage the encoder to capture learner-writing characteristics that are shared across datasets, without necessarily improving its alignment with the task-specific scoring conditions of any single downstream dataset, as we have already shown in Section 4.1. This suggests that continued pretraining on learner writing can help mitigate the performance degradation caused by cross-dataset transfer under few-shot adaptation, even when the benefit is asymmetric and more evident in some transfer settings than others.

4.3 Ablation Study

Table 3 presents the results of the ablation experiment. Specifically, the variant pretrained exclusively on B1–B2–C1 texts performs better than both the full EFCAMDAT variant and the A1–A2-only variant, suggesting that the more proficient subset of the pretraining corpus perhaps aligns better with the downstream datasets than the corpus as a whole or the beginner-level subset. These findings provide further support for our interpretation in Section 4.1 that the negative performance results of the domain-adapted model may be partly explained by a mismatch between the proficiency distribution of the pretraining corpus and that of the downstream datasets.

Nevertheless, all domain-adapted variants underperformed the base BERT model overall, with one exception on the FCE dataset, where the B1–B2–C1-only variant achieved a higher QWK score. This further supports our earlier interpretation that mismatches between EFCAMDAT and the downstream datasets extend beyond proficiency distribution, which are not addressed by the proficiency-based ablation alone. Taken together, these results suggest that the pretraining corpus was not sufficiently aligned with the downstream AES task and

Model	Setting	Transfer to FCE				Transfer to IELTS			
		QWK	RMSE	Spearman	Pearson	QWK	RMSE	Spearman	Pearson
DAPT BERT	In-domain	0.405	2.386	0.569	0.565	0.538	0.911	0.722	0.686
	Transfer	0.000	10.093	-0.170	-0.073	0.000	2.957	0.236	0.255
	Drop	0.405	7.707	0.739	0.638	0.538	2.046	0.486	0.431
BERT	In-domain	0.498	2.163	0.629	0.636	0.661	0.817	0.773	0.794
	Transfer	0.000	11.297	-0.085	0.001	0.000	4.217	0.032	0.074
	Drop	0.498	9.134	0.714	0.635	0.661	3.400	0.741	0.720

Table 2: Cross-dataset transfer results under the few-shot setting. DAPT BERT denotes bert-base-uncased further pretrained on EFCAMDAT. For each target dataset, we report in-domain performance, transfer performance, and the absolute performance drop. Lower RMSE indicates better performance, whereas higher QWK, Spearman, and Pearson indicate better performance.

Pretraining Data	RMSE	QWK	Sp.	Pe.
IELTS				
No EFCAMDAT	0.817	0.661	0.773	0.794
B1-B2-C1 only	0.860	0.567	0.753	0.731
A1-A2 only	0.936	0.542	0.745	0.684
Full EFCAMDAT	0.911	0.538	0.722	0.686
FCE				
No EFCAMDAT	2.163	0.498	0.629	0.636
B1-B2-C1 only	2.226	0.550	0.598	0.614
A1-A2 only	2.271	0.451	0.595	0.602
Full EFCAMDAT	2.386	0.405	0.569	0.565

Table 3: Ablation results on IELTS and FCE. “No EFCAMDAT” denotes the base BERT model without domain-adaptive pretraining. Sp.=Spearman and Pe.=Pearson. Best result for each dataset and metric is shown in bold.

that its broad learner language coverage may have limited its effectiveness.

5 Conclusion and Future Work

Our study investigated whether continued pretraining on an English learner corpus could improve automated essay scoring (AES) for English proficiency tests in two aspects: in-domain scoring performance and the mitigation of performance drop under cross-dataset transfer. To examine this, we continued pretraining BERT on the EFCAMDAT corpus and conducted two sets of experiments: baseline comparisons and cross-dataset transfer. The results showed that the domain-adapted BERT model performed worse on both downstream datasets when competing against baseline models and its own original base checkpoint. However, in the cross-dataset transfer experiment, the domain-adapted BERT generally exhibited a smaller transfer-induced performance drop than the original BERT model, with clearer evidence in one transfer direction and more modest evidence in the other. Although the benefits are not uniform across all settings, the transfer results indicate that continued pretraining on learner writing may be useful

in realistic low-resource settings involving transfer across English proficiency tests, where annotated essays are abundant for one test but limited for another.

Additionally, the ablation study revealed that pretraining on only higher CEFR proficiency texts (B1-B2-C1) yields better performance in comparison to pretraining on the full EFCAMDAT and beginner-level texts only, which may be the result of better alignment between the pretraining corpus and our downstream datasets. These findings suggest that there is likely no universally optimal pretraining corpus for AES on English proficiency tests. Instead, future work aiming to improve robustness on English proficiency assessment should consider selecting adaptation data that more closely matches the downstream writing tasks, especially in terms of proficiency range and communicative purpose.

Limitations

Our study has several limitations. First, continued pretraining was conducted on only one learner corpus, EFCAMDAT, so the findings may not generalize to learner-domain adaptation more broadly. Second, the adaptation corpus may not be sufficiently aligned with the downstream AES datasets in terms of proficiency distribution, genre, and communicative purpose, which may have contributed to the weak in-domain results. Third, only BERT was domain-adapted, so it remains unclear whether similar patterns would be observed for other pre-trained encoder architectures. Finally, the downstream evaluation was limited to two essay datasets and a specific few-shot transfer setting, which constrains the generalizability of the transfer findings.

Ethics Statement

We plan to make our code publicly available after review. However, access to EFCAMDAT is governed by a License Agreement and is restricted to academic, non-commercial research, with conditions protecting copyright. For this reason, we will not redistribute the corpus or any text-derived data artifacts with our code, including the model checkpoints, as it risks the regulated terms being violated beyond our control. Researchers wishing to reproduce our experiments must obtain access to EFCAMDAT through the official provider.

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A Reproducibility

To support reproducibility, we report the main hyperparameters used in continued pretraining and downstream fine-tuning.

The random seed was set to 42. The maximum sequence length was 512, and the masking probability was 0.15. The model was trained for 1 epoch with a per-device training batch size of 1 and a per-device evaluation batch size of 1. Gradient accumulation was set to 8, resulting in an effective batch size of 8. The learning rate was 5×10^{-5} ,

weight decay was 0.01, and warm-up was set to 5154 steps. A linear learning rate scheduler was used. Training was conducted in FP16 precision, and 5% of the corpus was reserved as a validation split.

For downstream fine-tuning, the learning rate was 2×10^{-5} . The per-device training batch size was 8, and the per-device evaluation batch size was 16. Fine-tuning was run for 5 epochs.